

IndoSBERT: Enhancing Indonesian Sentence Embeddings with Siamese Networks Fine-tuning

Kadek Denaya Rahadika Diana
School of Electrical Engineering and Informatics
Institut Teknologi Bandung
Indonesia
denayarahadika@gmail.com

Masayu Leylia Khodra
School of Electrical Engineering and Informatics
Institut Teknologi Bandung
Indonesia
masayu@staff.stei.itb.ac.id

Abstract—Sentence embeddings hold a crucial role in Indonesian NLP research, being applied in various domains such as essay scoring, summarization, text-to-image generation, text classification, and others. By this research, we present IndoSBERT, a modification of BERT that has been fine-tuned using the siamese network scheme inspired by SBERT. This model was fine-tuned with the STS Benchmark Dataset which was translated into Indonesian language. Our model can provide meaningful semantic sentence embeddings for Indonesian sentences. IndoSBERT achieved higher scores in Semantic Textual Similarity task compared to existing and widely used models, such as IndoBERT by IndoNLU and IndoBERT by IndoLEM.

Keywords—BERT, semantic textual similarity, sentence embedding

I. INTRODUCTION

In NLP, embeddings are one of the methods used to represent human language (in the form of words or sentences) in numerical form. By transforming text into meaningful contextual numerical vectors, words or sentences can be accurately represented by these vectors. This allows us to utilize these numerical representations for various purposes such as text classification, sentiment analysis, machine translation, named entity recognition, and document clustering.

BERT [1] is a transformer-based model widely used in the field of NLP, mainly due to its ability to create high-quality embeddings for a given token. While BERT is effective in capturing word-level contextual information, the objective of the training process is not explicitly to train sentence-level semantics. As a result, it may not fully capture the overall meaning or context of an entire sentence, limiting its ability to produce reliable sentence embeddings.

SBERT [2] is a modified version of BERT that is fine-tuned to enable BERT to provide meaningful semantic information. SBERT is fine-tuned using the siamese network mechanism. The siamese network architecture involves using two identical neural networks that share weights and parameters to process two different inputs simultaneously. This setup allows SBERT to learn better sentence-level representations by focusing on tasks related to measuring semantic similarity or dissimilarity between sentence pairs.

Fine-tuning SBERT is conducted by using Semantic Textual Similarity Benchmark Dataset [3]. This dataset contains pairs of sentences along with their corresponding similarity scores. A score of 0 indicates that the two sentences have no semantic meaning connection at all, while a score of 5 indicates that the sentences are closely related in meaning.

To create a model capable of providing high-quality sentence embeddings for Indonesian text, we utilize a machine-translated Indonesian Semantic Textual Similarity Benchmark Dataset [4].

Currently, there are several existing IndoBERT models, such as IndoBERT by IndoLEM [5] and IndoBERT by IndoNLU[6]. Additionally, there are already some multilingual pretrained models available that can be used for the Indonesian language. However, it is important to note that these models were not explicitly trained for the Semantic Textual Similarity (STS) task by using Indonesian sentences. We also found a pretrained Indonesian sentence BERT, but we observed that this model has a lower STS test dataset score compared to the pretrained multilingual model provided by sentence-transformers. Thus, we found out that this BERT model does not provide a good score in the semantic textual similarity task for the Indonesian language.

In this research, we contribute by providing a fine-tuned BERT model using the STS Benchmark Dataset in the Indonesian language, which we referred to as IndoSBERT which can provide meaningful sentence embeddings for Indonesian sentences and achieved higher score in Semantic Textual Similarity task compared to existing and widely used IndoBERT models.

The rest of the paper is organized as following. Section II presents the related work that supports this research. Section III describes the process of creating and training IndoSBERT. Section IV presents the research and discussion of the IndoSBERT model obtained from the preceding sections. Section V provides the evaluation results on our test dataset, as well as an in-depth analysis of the performance and findings. In Section VI, we also provide a comprehensive conclusion of our research, outlining the key findings and key takeaways of our study. Additionally, we offer valuable recommendations for further research in this domain to explore potential improvement in the field of sentence embeddings for Indonesian language NLP.

II. RELATED WORKS

Sentence embeddings have a crucial role in Indonesian NLP research, being applied in various domains such as essay scoring, summarization, text-to-image generation, text classification, and others. However, some studies in the Indonesian NLP domain have not utilized sentence embeddings that have been specifically trained to provide robust semantic representations. Sentence embeddings has been used for the task of essay scoring in Indonesian language essays [7]. But, in that research paper, the author used pretrained embeddings that were not specifically trained for the Indonesian language. Another study creates sentence

embeddings by utilizing word embeddings and RNN (Recurrent Neural Network), as in [8]. These sentence embeddings are then fed into a graph convolutional network to generate an automatic summarization system.

A research utilizes sentence embeddings to generate Text-to-Image Synthesis in the Indonesian language using SentenceBERT and FastGAN [9]. IndoBERT is used as the basis for SentenceBERT. However, its sentence embeddings are not explicitly trained to compute semantic similarity between two sentences using a dataset built by humans.

In addition, sentence embeddings can also be utilized for text classification in the Indonesian language. The study [10] creates sentence embeddings by performing word vector averaging from FastText [11]. However, using word vectors to form sentence embeddings cannot generate vectors that provide a good semantic representation of sentences [2].

Currently, there are several research efforts developing BERT-based models specifically for the Indonesian language. IndoLEM has developed IndoBERT [5], trained on 220 million words using a masked language model training framework. IndoNLU has created their version of IndoBERT model [6], which is trained using the Indo4B corpus, aiming to evaluate 12 downstream tasks. These two IndoBERT models were not explicitly trained to produce semantically meaningful sentence embeddings. Our test results for these two models indicate that even when trained using Indonesian dataset and widely used in NLP domain for Indonesian language, these two models still cannot provide satisfactory performance in semantic textual similarity for the language.

SBERT is a sentence embedding model that utilizes a siamese network to fine-tune the BERT model [2]. The goal of SBERT is to create two embedding vectors that are close together when two sentences have similar semantic meanings. The model is finetuned by using siamese network as shown in Fig 1.

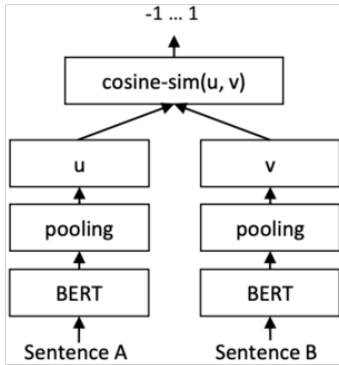


Fig. 1. Siamese network architecture that is used to make SBERT [2].

Currently, we have found SBERT models that have been trained for the Indonesian language [12]. However, when using the STS Benchmark test dataset in the Indonesian language, we observe that the Spearman Rank correlation scores obtained are still not higher than the existing pretrained multilingual SBERT model.

Through this research, we proposed IndoSBERT, an SBERT model specifically designed for the Indonesian language, capable of generating effective sentence representations. We utilize a pretrained Indonesian BERT model [5-6] as the foundation for fine-tuning. Our aim is to

provide this model to be utilized in a wide range of downstream tasks within Indonesian NLP.

III. PROPOSED MODEL

In this section, we will explain how we obtained the best configuration for IndoSBERT. In creating IndoSBERT, we utilized a siamese network inspired by the training mechanism used in SBERT. To construct this siamese network, we leveraged several pretrained IndoBERT models, such as IndoBERT by IndoLEM and IndoBERT by IndoNLU.

Unlike the architecture provided in SBERT, we also added an optional additional layer, namely the dense layer, after the pooling process of the BERT layer. This was done to enhance the model's ability to absorb semantic information from sentences. Fig. 2 displays the siamese network architecture that we used to fine-tune the BERT models for Indonesian language.

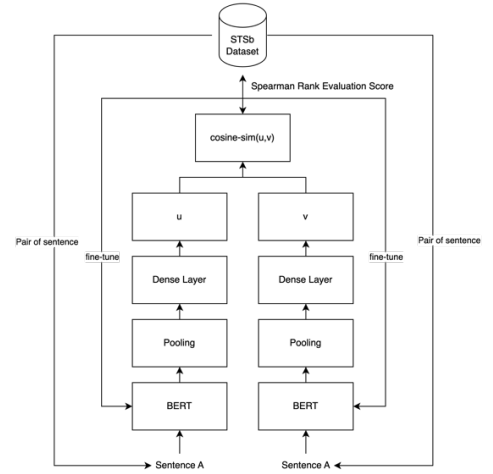


Fig. 2. Siamese network architecture that is used to make IndoSBERT.

We utilized the Semantic Textual Similarity Benchmark Dataset, which has been machine-translated into the Indonesian language and made available open source [4]. Two sentences from each row in this dataset will be fed into the provided siamese network. Subsequently, the siamese network will undergo the fine-tuning process for the BERT input model.

The two resulting sentence embeddings from each part of the siamese network will be used to calculate the cosine similarity score between the two sentences. Subsequently, this cosine similarity score will be employed to find the Spearman Rank by comparing the siamese network's cosine similarity values with the label values of the two sentences in the STS Benchmark dataset.

Fig 3 displays the distribution of human scores in the overall machine-translated STS benchmark dataset. The graph indicates a left-skewed distribution, indicating a larger proportion of data with higher scores. This dataset is divided into 10,321 training data and 2,608 test data. For the 10,321 training data, we utilize 20% of it as validation data during the learning process.

Later on, we will use the test dataset to assess the performance of our best model obtained from the learning and validation process and its generalizability and real world applicability. Additionally, we will provide an analysis of the

model's effectiveness, along with the inference results on the test dataset.

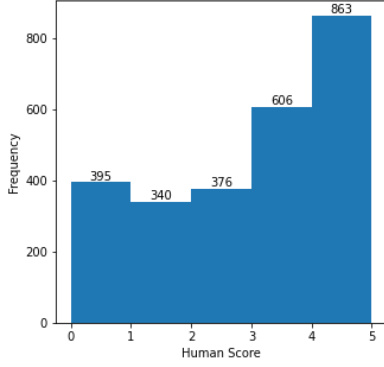


Fig. 3. Distribution of human scores in the machine-translated STS Benchmark Indonesian Dataset.

Spearman Rank is a statistical measure used to assess the correlation between two ranked variables. In the context of evaluating the performance of the siamese network in this research, Spearman Rank is employed to determine the correlation between the predicted cosine similarity values and the human-assigned similarity scores in the STS Benchmark dataset. By calculating the Spearman Rank, we can quantitatively assess how well the siamese network's predicted similarities align with the human-assigned similarity labels. A higher Spearman Rank score indicates a stronger correlation and better alignment with human judgments, suggesting that the siamese network is more effective in capturing the semantic similarity between sentences. This information aids in evaluating and comparing the performance of different models in the context of the STS Benchmark dataset.

IV. RESULTS AND DISCUSSION

To create IndoSBERT, we used three base BERT models. The first model is IndoBERT-IndoLEM `indolem/indobert-base-uncased`. Additionally, we utilized IndoBERT-IndoNLU `indobenchmark/indobert-base-p1` and `indobenchmark/indobert-large-p1`. In addition to that, we employed various multilingual BERT models as our base models, such as `distiluse-base-multilingual-cased-v2` and `paraphrase-multilingual-MiniLM-L12-v2`. Every model that we used is available open source and can be accessed on Hugging Face website.

The experiments were conducted using the Google Colab environment with a V100 GPU. The dataset was divided into three parts: train, dev, and test datasets. The training process was carried out with 50 epochs and a batch size of 16. The optional additional layer we utilized is a dense layer with an input vector of length 256 and an output vector of the same length. The activation function used is tanh, and the loss function employed is cosine similarity loss.

Fig. 4 displays the Spearman Rank Correlation Score values for models fine-tuned using this dataset. The one labeled as `'_dense_trained'` is the model fine-tuned using a siamese network with an additional dense layer. It can be seen that `indobenchmark/indobert-large-p1`, when fine-tuned without using an additional dense layer, provides better scores on the validation dataset compared to other IndoBERT models. However, the SBERT model fine-tuned from the IndoBERT model still produces inferior results compared to

the multilingual `paraphrase-multilingual-MiniLM-L12-v2` model fine-tuned with STS.

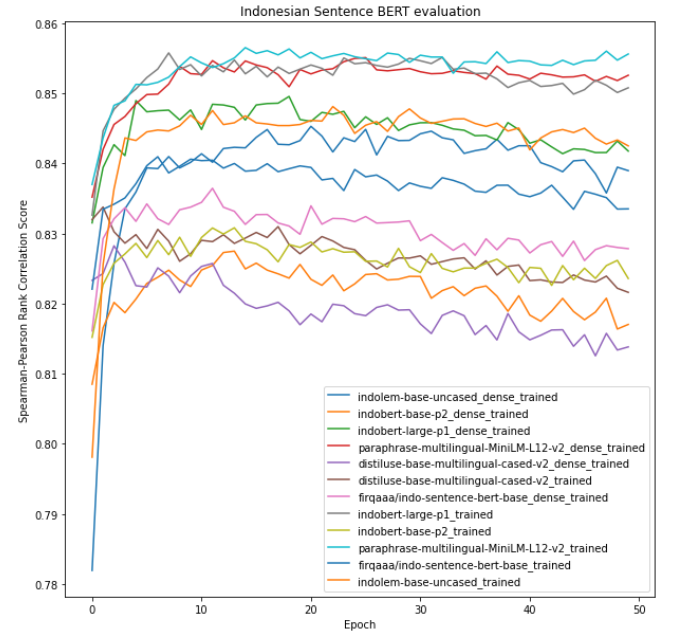


Fig. 4. Spearman rank correlation scores of fine-tuned models on the dev dataset.

V. EVALUATION

In addition to evaluating our fine-tuned models, we also conducted tests on non-fine-tuned models to compare their performance with our fine-tuned model. Afterward, we attempted to apply both the fine-tuned and non-fine-tuned models to perform the Semantic Textual Similarity task on the test data that we prepared earlier, as shown in Table I. This comparison allowed us to gain insights into the impact of fine-tuning on the models' ability to capture semantic similarities between sentences and to assess the overall effectiveness of our approach.

TABLE I. MODEL EVALUATION TO STS BENCHMARK TEST DATASET

Model	Spearman Rank*
<i>Not trained for STS</i>	
<code>indolem/indobert-base-uncased</code>	0.439
<code>indobenchmark/indobert-large-p1</code>	0.550
<code>indobenchmark/indobert-base-p1</code>	0.552
<i>Pretrained multilingual model (50+ languages)</i>	
<code>distiluse-base-multilingual-cased-v2</code>	0.715
<code>paraphrase-multilingual-MiniLM-L12-v2</code>	0.775
<i>Trained for STS</i>	
<code>firqaai/indo-sentence-bert-base-trained</code>	0.841
<code>distiluse-base-multilingual-cased-v2-dense-trained</code>	0.828
<code>paraphrase-multilingual-MiniLM-L12-v2-trained</code>	0.856
<code>indobenchmark/indobert-base-p2-dense-trained</code>	0.827
<code>indolem/indobert-base-uncased-dense-trained</code>	0.845
<code>indobenchmark/indobert-large-p1-trained (IndoSBERT)</code>	0.856

Based on these results, we concluded that indobenchmark/indobert-large-pl-trained, our fine-tuned indobenchmark/indobert-large-pl model using translated STS Benchmark Dataset, achieved the best performance among the models, along with paraphrase-multilingual-MiniLM-L12-v2-trained, demonstrating a strong capability in capturing semantic similarities between sentences.

When we apply IndoSBERT model to the entire translated STS Benchmark dataset (combining train, dev, and test datasets), we found that the model is capable of producing score distributions that align with the scores generated by humans, as shown in the following Fig 5. We can observe that the model's inference distribution also follows the traits of the overall dataset, such as its skewness. The boxplot in Fig 6 also indicates that the distribution of both the original and inferred data is similar. Notably, the mean and median are found in the same region of the dataset.

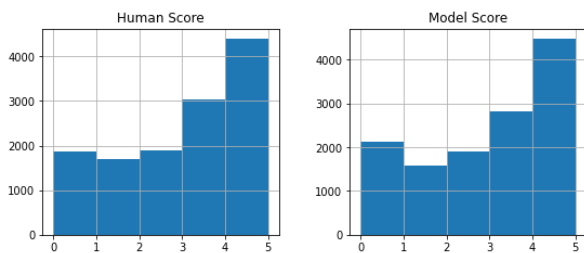


Fig. 5. Histogram of score distribution of human generated (left) and IndoSBERT model's (left) to the overall STSb dataset.

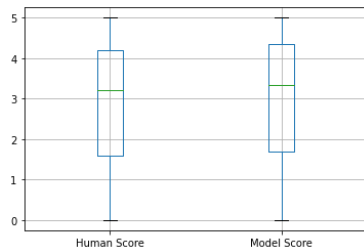


Fig. 6. Boxplot of score distribution of human generated (left) and IndoSBERT model's (left) to the overall STSb dataset.

However, when the model is applied to infer the test dataset, it is noticed that the model tends to produce higher cosine similarity values, as shown in the following Fig 7 and Fig 8. This can be observed from the distribution of average scores resulting from the inference, which shows fewer instances in the 0-1 range. This indicates that the model tends to infer higher values for the similarity between sentences.

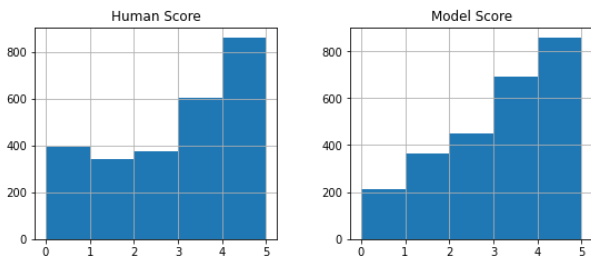


Fig. 7. Score distribution of human generated (left) and IndoSBERT model's (left) to the STSb test dataset.

The following Table II shows an example comparison between human-generated similarity scores and the similarity

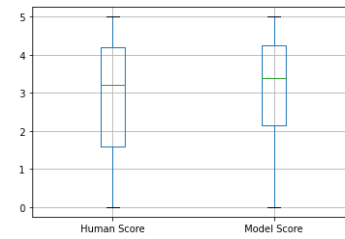


Fig. 8. Score distribution of human generated (left) and IndoSBERT model's (left) to the STSb test dataset.

scores generated by our model. The sentences are written in Indonesian, with their English translations provided. Our model demonstrates good results in inferring the sentence pair similarity.

Table II indicates that the model is capable of predicting two non-related sentence pairs, as shown in the latest row. The first sentence discusses food, while the second sentence pertains to music. Despite both sentences having syntactic similarity, the model effectively captures that their semantic and contextual meaning are unrelated.

TABLE II. SENTENCE PAIR SIMILARITY SCORE (HUMAN-GENERATED VS INDOBERT-GENERATED)

Sentence 1	Sentence 2	Human score	Model score
Seperti yang sudah saya jelaskan pada bacaan kedua, ada krisis yang mendasari amandemen arahan ini.	Seperti yang telah saya katakan dalam bacaan kedua, krisis merupakan dasar dari amandemen arahan ini.	4.75	4.83
<i>As I already explained during second reading, there is a crisis underlying this directive amendment.</i>	<i>As I have already explained in second reading, is at the root of this directive.</i>		
Sebuah sepeda motor diparkir di dekat dinding yang ditutupi dengan pemandangan grafiti pemandangan kota.	Sebuah sepeda motor diparkir oleh mural sebuah kota di malam hari	3.40	4.31
<i>A motorcycles is parked near a wall covered with a graffiti scene of a cityscape.</i>	<i>A motorcycle is parked by a mural of a city in the evening.</i>		
Apakah ada cara untuk mengetahui apakah saya memerlukan visa transit untuk singgah di Inggris?	Saya memiliki Layover di Jepang. Akan tinggal selama sebulan, Apakah saya memerlukan Visa Transit?	1.0	0.84
<i>Is there a way to find out if I need a transit visa for a layover in the UK?</i>	<i>I have a Layover in Japan, Going to stay for a month, Do I need a Transit Visa?</i>		
Apa perbedaan antara shawarma dan gyros?	Apa perbedaan antara portamento dan glissando?	0	0.10
<i>What is the difference between shawarma and gyros?</i>	<i>What is the difference between portamento and glissando?</i>		

Based on the inference results from the above testing data, we found that the largest difference range between human score and model score is 3.37. The sentence pair with the

highest human-machine score difference is shown in the Table III. We found out that the sentence pair with the worst score discrepancy is due to a translation error. The translated sentence appears to have no meaning in Indonesian. This is because word ‘can’ is an English homonym word, leading to the translation error. The translated sentence of a homonym results in a nonsensical sentence in the Indonesian language.

TABLE III. SENTENCE PAIRS WITH THE HIGHEST HUMAN-MACHINE SCORE DIFFERENCE

Sentence 1	Sentence 2	Human score	Model score
<i>Why do you need to peel peaches to can them? (original)</i>	<i>How to peel peaches? (original)</i>	1.0	4.37
Mengapa Anda perlu mengupas buah persik agar bisa? (translated)	Bagaimana cara mengupas buah persik? (translated)		

Furthermore, we also observed that out of the total 2580 samples, 2011 of them were predicted to have a score difference of less than 1. This indicates that the model's predictions were relatively close to the human scores for a considerable portion of the dataset. The detailed breakdown of data points below specific difference thresholds between the human score and model score can be seen in Table IV.

TABLE IV. COUNT OF DATA POINTS HAVING A DIFFERENCE BELOW A SPECIFIC THRESHOLD.

Threshold	# Data / Total test data
< 1	2054 / 2580
< 0.1	332 / 2580
< 0.01	47 / 2580
< 0.001	27 / 2580

These findings suggest that the model performs well on a significant portion of the test dataset, accurately predicting scores close to the human-annotated score. It shows the model's capability to capture semantic similarities between sentences, especially for cases with minimal score differences.

In addition, we also discovered several errors in the STS dataset concerning machine translation, as shown in the Table V. The sentence pairs convey different meanings in English. However, when translated to Indonesian via machine translation, they return the same sentence, resulting in an imprecise value in the human score.

TABLE V. MACHINE TRANSLATION ERRORS IN INDONESIA STS DATASET

Sentence 1	Sentence 2	Human score	Model score
<i>World's oldest man dies at 116 (original)</i>	<i>World's oldest ever man dies aged 116 (original)</i>	4.6	5
Pria tertua di dunia meninggal pada usia 116 (translated)	Pria tertua di dunia meninggal pada usia 116 (translated)		
<i>Van Orden Report (A5-0241/2000) (original)</i>	<i>Report Van Orden (A5-0241 / 2000) (original)</i>	4.75	5

Laporan Van Orden (A5-0241 / 2000) (translated)	Laporan Van Orden (A5-0241 / 2000) (translated)		
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VI. CONCLUSION AND RECOMMENDATION

We presented a sentence embedding model specifically trained for Indonesian language called IndoSBERT, which was fine-tuned on the machine translated Semantic Textual Similarity Benchmark Dataset. IndoSBERT utilizes the siamese network architecture to learn better sentence-level representations and measure semantic similarity or dissimilarity between sentence pairs.

Compared to existing IndoBERT models, pretrained Indonesian language sentence embedding models, as well as some multilingual pretrained models, our fine-tuned models achieved higher scores in the Semantic Textual Similarity task, providing more meaningful sentence embeddings for Indonesian sentences. This indicates that IndoSBERT is a promising model for capturing semantic information in Indonesian text and could be beneficial for various natural language processing applications. Our model IndoSBERT can be accessed via <https://huggingface.co/denaya/IndoSBERT-large>.

For future work, we recommend the reader to explore the capability of IndoSBERT in a wider range of natural language processing tasks beyond the Semantic Textual Similarity task. We believe that by investigating its performance in other tasks like text classification, sentiment analysis, and document summarization might provide a much more insights of the IndoSBERT capability and effectiveness.

Additionally, we also encourage the reader to employ this IndoSBERT as a base to do downstream tasks and making it more useful for real world tasks in various industries. Continuous efforts in evaluating and refining IndoSBERT will contribute to its potential as a reliable and efficient model for Indonesian language understanding in the field of natural language processing.

As of the present, we are utilizing the machine-translated Semantic Textual Similarity Dataset, which has resulted in several errors, as evident from our test evaluation. It has become apparent that relying solely on a machine-translated dataset is insufficient, given the presence of errors. We firmly believe that by creating a more extensive Indonesian STS Benchmark Dataset, enriched with a larger number of data points, and complemented by human annotation, we can significantly enhance the performance and capabilities of our Indonesian Language model in the future.

REFERENCES

- [1] D. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, vol. 1, pp. 4171-4186, 2019.
- [2] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: Findings, 2020, pp. 4292-4302.
- [3] "Papers with Code - Semantic Textual Similarity 2012-2016." Available online: <https://paperswithcode.com/dataset/semantic-textual-similarity-2012-2016> (Accessed July 27, 2023).
- [4] A. Izzan. (2020). STS Indonesia. [Online]. Available: <https://github.com/ahmadizzan/sts-indo>

- [5] F. Koto, A. Rahimi, J. H. Lau, and T. Baldwin, "IndoLEM and IndoBERT: A Benchmark Dataset and Pre-trained Language Model for Indonesian NLP," in Proceedings of the 28th International Conference on Computational Linguistics, Barcelona, Spain (Online), Dec. 2020, pp. 757-770.
- [6] B. Wilie, K. Vincentio, G. I. Winata, S. Cahyawijaya, X. Li, Z. Y. Lim, S. Soleman, R. Mahendra, P. Fung, S. Bahar, and A. Purwarianti, "IndoNLU: Benchmark and Resources for Evaluating Indonesian Natural Language Understanding," in Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing, Suzhou, China, Dec. 2020, pp. 843-857.
- [7] R. A. Rajagede, "Improving Automatic Essay Scoring for Indonesian Language using Simpler Model and Richer Feature," Kinetik: Game Technology, Information System, Computer Network, Computing, Electronics, and Control, vol. 6, no. 1, 2021.
- [8] G. Garmastewira and M. L. Khodra, "Summarizing Indonesian news articles using graph convolutional network," Journal of Information and Communication Technology, vol. 18, no. 3, pp. 345-365, 2019.
- [9] M. R. S. Mahadi and N. P. Utama, "Indonesian Text-to-Image Synthesis with Sentence-BERT and FastGAN," 2023. [Online]. Available: arXiv:2303.14517 [cs.CV].
- [10] A. Amalia, O. S. Sitompul, E. B. Nababan, and T. Mantoro, "An Efficient Text Classification Using fastText for Bahasa Indonesia Documents Classification," in 2020 International Conference on Data Science, Artificial Intelligence, and Business Analytics (DATABIA), Medan, Indonesia, 2020, pp. 69-75, doi: 10.1109/DATABIA50434.2020.9190447.
- [11] "fastText." [Online]. Available: <https://fasttext.cc/>. [Accessed: 20-Sep-2019].
- [12] Arasyi, F. (2022). Indo-Sentence-BERT: Indonesian Sentence BERT for Semantic Similarity. [Online]. Available: <https://huggingface.co/firqaaa/indo-sentence-bert-base>